

Module 12: Physical Transmission Media & Multiplexing

AKTU BCS603 Focus: Guided media specs, Optical Fiber TIR physics, Wireless frequency propagation, and Multiplexing architectures (FDM, WDM, Synchronous vs Statistical TDM).

Physical Transmission Media (Guided vs Unguided) & Multiplexing (FDM, WDM, TDM)

A. Guided Media (Wired Conduits)

1. Twisted Pair (UTP / STP):

- Twisting cancels electromagnetic interference (EMI) & crosstalk.
- Cat 5e (100 MHz, 1 Gbps), Cat 6 (250 MHz, 10 Gbps up to 55m). RJ-45.

2. Coaxial Cable:

- Center copper core, dielectric insulator, metallic braided shield, jacket.
- 50 Ω Baseband (digital), 75 Ω Broadband (analog cable TV). BNC connector.

3. Optical Fiber (Glass / Silica):

- Total Internal Reflection (TIR): Core index n_1 > Cladding index n_2 .
- Single-mode (narrow core, laser, 100s km) vs Multi-mode (LED, short haul).
- Immense bandwidth, immune to EMI, zero spark hazard, lowest loss.**

B. Unguided Media (Wireless Propagation)

1. Radio Waves (3 kHz to 1 GHz):

- Omnidirectional antenna (broadcasts in all directions).
- Penetrates walls & buildings easily. FM Radio, AM, TV broadcast.

2. Terrestrial & Satellite Microwaves (1 to 300 GHz):

- Strictly Unidirectional line-of-sight propagation (dish antennas).
- Blocked by obstacles; requires repeater towers every 40-50 km.
- Geosynchronous Satellite uplink (6 GHz) & downlink (4 GHz).

3. Infrared (300 GHz to 400 THz):

- Strictly Short-range closed-room line-of-sight.
- Cannot penetrate walls (high security). TV remotes, wireless mice.

Multiplexing: Combining Multiple Signals onto a Single Physical Channel

FDM (Frequency Division)

- Analog signals divided into frequency bands.
- Separated by Guard Bands to avoid overlap.
- E.g., Radio broadcast, Cable TV channels.
- Channel bandwidth >> Signal bandwidth**

WDM (Wavelength Division)

- Optical equivalent of FDM for Fiber cables.
- Combines different wavelengths (colors of light).
- Multiplexer/Demultiplexer uses optical prism.
- DWDM delivers Terabits/sec over 1 fiber!**

TDM (Time Division)

- Digital signals share full bandwidth in time slots.
- Synchronous TDM: Pre-allocated time slots.
- Statistical TDM: On-demand dynamic slots.
- Eliminates wasted idle time slots!**

1. Single-Mode vs Multi-Mode Fiber Comparison

Feature	Single-Mode Fiber (SMF)	Multi-Mode Fiber (MMF)
Core Diameter	8 to 10 μm (Extremely thin)	50 to 62.5 μm (Wide core)
Light Source	Laser Diodes (Infrared 1310/1550 nm)	LEDs (Light Emitting Diodes 850 nm)
Modal Dispersion	Zero (Single ray travel)	High (Multiple rays bounce at different angles)
Distance Range	Up to 100+ km without amplification	Up to 2 km (Campus / Building backbones)

2. Synchronous TDM vs Statistical TDM

- Synchronous TDM:** Frame size is fixed. If host A is silent, slot A travels empty across the WAN. Efficiency drops when traffic is bursty.
- Statistical TDM:** Slots are dynamically addressed. Only stations with data in buffer are awarded slots. Requires addressing overhead in each slot, but maximizes throughput.

AKTU Exam Question: "Discuss different physical layer transmission media" me Guided (Twisted pair, Coax, Fiber) aur Unguided (Radio, Microwave, IR) dono ka neat architectural comparison table zaroor banayein!